

JOSHUA CLINTON

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Penn State computer science senior graduating December 2026, seeking full-time software and firmware engineering roles starting early 2027, especially where AI is breaking new ground.

EDUCATION

Pennsylvania State University - University Park, PA

Bachelor of Science: Computer Science, *Expected 12/2026*

- **Minors:** Computer Engineering, Engineering Leadership Development, Cybersecurity Computational Foundations

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++, C, Java, PHP, MATLAB, Shell, HTML

Technologies: Git, React, Node.js, Flask, Docker, Jenkins, Xilinx Vivado, Embedded Systems, Claude Code

AI / Machine Learning: PyTorch, NumPy, Computer Vision, ResNet, YOLO, OCR, Model Training

WORK EXPERIENCE

Software Engineer Intern – *Northrop Grumman* | June 2026 – August 2026

- Contributed to an **Agile, sprint-based software development team**, participating in sprint planning, daily stand-ups, and peer code reviews.
- Developed and debugged **low-level C++ for system-level software** interfacing with FPGA-based hardware.

Software Engineer Intern – *B. Braun Medical Inc.* | May 2025 – May 2026

- Built a **Python-based AI computer vision system** to inspect medical vials and syringes, detecting errors in NDC codes, expiration dates, lot numbers, and syringe volumes. **Named inventor on the pending patent** covering the inspection system.
- Designed and trained **custom neural networks** using ResNet and YOLO architecture, integrating EasyOCR for text extraction, achieving accurate, real-time defect detection.
- **Modernized** a legacy C# label-printing application into a fully customizable **web-based platform** with a PHP backend, enabling secure data handling and integration with existing systems.
- **Cut labeling setup time** from ~10 minutes to just a few seconds through a one-click print workflow, improving production efficiency across multiple facilities.

AI Trainer, Coding & Software – *Handshake AI* | June 2024 – Present

- Build benchmark datasets that test AI coding models, turning real GitHub pull requests into verifiable tasks across Go, Python, and Rust repositories (**Project Helix**, in collaboration with Anthropic).

ACCOMPLISHMENTS AND PROJECTS

Pipelined MIPS CPU

Designed and simulated a 5-stage pipelined processor using Verilog and Xilinx Vivado with hazard detection and forwarding logic. Validated CPU performance using custom assembly test programs and waveform analysis.

Monte Carlo Path Tracer

From-scratch C++17 path tracer with physically based materials, BVH acceleration (6.24× faster than linear scan), and ReSTIR-style resampled direct lighting.

Neural Car

A car that teaches itself to drive randomly generated tracks; the neural network and genetic algorithm are written from scratch in NumPy, with no PyTorch or autograd.

ACM Coding Competition

Secured 2nd place in the ACM coding competition hosted by DeSales University, demonstrating problem-solving skills under pressure.